

Deep Reinforcement Learning

Actor-Critic Algorithms

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Content

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- Two alternative derivations of the policy gradient theorem
 - Sampling with reduced variance
 - Bottom-up derivation via the Q-function
- Advantage functions

Where are we?

Chapter	Topic	Content
	Basics & tabular methods	
1-5	Bandits, MDPs, Dynamic Programming, Monte Carlo, TD Learning	RL basics in finite dimensions
	Deep-learning-based methods	
6	Brief introduction to deep learning	The basics for what comes next
7	Value function approximation	Value estimation with function approximation
8	Deep Q-learning	Q-learning with neural networks
9	Policy gradients	Direct optimization of the policy
10	Actor-critic algorithms	Improved policy gradients via value functions
11	Advanced algorithms	
	Model-Based Control	
	Advanced Topics	

References